

Compact Photonic Model Based on Coupled-Mode Theory for Nonlinear Interactions in Electronic-Photonic Co-Simulation

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Abstract: This paper demonstrates a Verilog-A compact photonic model based on coupled-mode theory for four-wave mixing (FWM) to present a general framework and methodology for modeling nonlinear interactions in electronic-photonic co-simulation. The model is compatible with existing EDA platforms and can support rapid electronic-photonic co-simulation. It avoids describing the complicated physical process of the FWM and provides an easy way for system designers to monitor the dynamics of the critical optical parameters, thus accelerating the co-design and co-optimization of the electronic-photonic hybrid systems incorporating FWM. Meanwhile, the model is not tailored for a particular photonic device but can be applied to different photonic devices as a fundamental component. The model framework and modeling methodology presented in this paper are expected to be further applied to the modeling of other optical nonlinear interactions. The simulation results of the proposed model agree well with both numerical and experimental results. A closed-loop electronic-photonic co-simulation example using the proposed model is also carried out successfully.

1. Introduction

Electronic-photonic heterogeneous-converging integrated circuits (EPHICs) are widely accepted as a promising technology in the post-Moore era [1, 2]. By sufficiently combining the strengths of EICs and PICs, many impending challenges in different fields, such as data transmission and processing, are expected to be addressed. Since the size and complexity of EPHICs have increased over the years, the need for electronic-photonic co-simulation and co-design becomes imminent.

However, some challenges remain. Many pieces of specialized software, such as Lumerical FDTD [3] and MODE [4], etc., are widely used to simulate photonic components at the physical device level. While these programs can provide the most accurate results, they are too computationally intensive to support the co-simulation of hundreds of devices, especially over long timescales [5]. Some system-level simulation tools based on MATLAB [6, 7] and programming languages [8, 9], as well as simulators such as OptiSpice [10, 11], Synopsys OptSim [12], and Lumerical INTERCONNECT [13] are also developed. These programs provide a platform for users to simulate and optimize their work from the system perspective. However, most of them are written as standalone software. They are difficult to adapt to existing circuit design and layout infrastructures, especially with the transistor SPICE or Spectre models provided

in standard process design kits (PDKs) [5]. While some of them do offer simulation interfaces with electronic design automation (EDA) tools, they simulate the electronic and photonic parts separately and idealize their interactions [5, 14, 15]. Cadence and its partners are currently leading the way. They implement the simulation and optimization of photonic circuits within the Cadence Virtuoso [16] environment by co-working with the Lumerical INTERCONNECT. The EDA and PDA simulators can operate in lockstep, and the electrical and optical domains can be updated as they interact during each simulation time step [17]. However, there are still some miles to go. Firstly, the Lumerical INTERCONNECT does not currently provide off-the-shelf models for optical nonlinear effects. Such simulation entails first designing the particular device and then extracting the scattering parameters in the Lumerical FDTD. It is quite a time-consuming and unfriendly process for circuits and systems designers with little optical expertise, especially when the photonic devices have complicated structures, which limits the widespread applications of this approach. Moreover, the devices' S-parameter matrix will change and must be reextracted if the ambient temperature or other relevant conditions vary slightly. Therefore, while the given scattering-parameter model can be adapted to many open-loop co-simulation scenarios, such as the optical signal modulation, it does not yet fulfill the needs of closed-loop co-simulation with continuous parameter variations, especially for systems with considerable nonlinear effects.

Many behavioral macro models for photonic devices based on Verilog-A [14, 15, 18–22] or SPICE [23–25] are also proposed to perform the pre-silicon validation of photonic components along with electronic circuits entirely within the EDA environment. They have been proven effective and efficient and can be used by IC designers to design EPHICs on mature EDA platforms such as Cadence Virtuoso [14, 18, 21, 25–27]. However, modeling for photonic components in the EDA environment is still in the development stage, and some issues have not been fully considered, such as nonlinear interactions between different wavelengths of light. For this reason, systems based on nonlinear interactions, including wavelength conversion and optical parametric amplification, etc., cannot be co-simulated entirely on the EDA platforms before, which limits the accuracy and the scope of application of the existing models.

To solve this problem, we present a Verilog-A compact photonic model for FWM supporting electronic-photonic co-simulation in this paper. It is a general model and can be further used to build more complex but practical photonic devices on EDA platforms, thus being applied to the co-simulation and co-design of various systems based on the FWM effect. Other nonlinear interaction are expected to be further modeled based on the framework and methodology presented in this paper. Moreover, the framework is general and models based on it can also be implemented in SPICE. The remainder of this paper is organized as follows: Section 2 reviews the existing photonic models in the literature. Section 3 presents the limitations of the previous framework and the proposed model's framework and principle. Section 4 validates the proposed model with simulation and measurement results. A closed-loop electronic-photonic co-simulation example is presented in this section as well. Section 5 draws the conclusion.

2. Existing Compact Photonic Models

The FWM phenomenon is one of the essential nonlinear optical interactions. It has been explored to realize wavelength conversion, phase conjugation, and photon pair generation, facilitating areas from nonlinear optical signal processing (NOSP) [28] to quantum optics [29]. However, like many other nonlinear optical interactions, it has yet to be fully considered in existing models.

Existing models can be divided into device-specific models [18–20, 23, 31–33] and general models [14, 15, 21, 24, 25, 30]. For the former, models for different photonic devices are reported, including micro-ring modulators (MRM) [18, 23], Mach-Zehnder modulators (MZM) [19], lasers [20, 31], and photodetectors (PD) [32, 33]. These models target specific device structures and aim to address the exact issue. Remodeling is inevitable when device structure changes or new needs arise. Since they were not initially designed for applications based on FWM effect,

Table 1. Comparison with Existing Photonic Models.

	[7] PR'18	[23] PR'19	[30] JLT'17	[24] JLT'22	[14] OE'15	[25] TCAS-I'24	This Work
Platform	MATLAB	EDA	EDA	EDA	EDA	EDA	EDA
Model Type	Passive Device	MRM	Optic-link	Passive Device	General Model	General Model	General Model
Modeling Principle	Scattering Parameter	Analytic	Analytic	Scattering Parameter	Analytic	Analytic	Analytic
Description Approach	MATLAB	SPICE	Verilog-A	SPICE	Verilog-A	SPICE	Verilog-A
Co-simulation		✓	✓		✓	✓	✓
Baseband Equivalence	✓			✓	✓	✓	✓
Bi-directional Transmission				✓	✓	✓	✓
Nonlinear Interactions							✓
Completeness	✓	✓	✓✓	✓✓	✓✓✓	✓✓✓	✓✓✓✓

they cannot be used directly without additional work.

There are also efforts to build general models on the EDA platform. [30] presents a library containing different kinds of photonic devices to enable simulation, validation, and optimization of the full optical link. However, only propagation loss is considered in its waveguide model. For this reason, it fails to meet our requirements for co-simulation and co-design of systems involving the FWM effect. A scattering-parameter model is proposed in [24]. For passive photonic devices, it matches well with measurements and can theoretically characterize almost all nonlinear effects. However, closed-loop electronic-photonic co-simulation requires photonic models that describe the physical properties under continuous parameter variations such as temperature. Since the scattering parameters of active photonic devices vary in different states, capturing device properties under continuous parameter variations is impossible with this approach. For this reason, it is inappropriate for systems with tight interplay between electronic and photonic components. [14] proposes decomposing complex photonic devices into fundamental components (waveguides, couplers, heaters, light sources, etc.). By modeling these fundamental components with analytic physical equations and using them to reconstruct complex photonic devices such as MRM and MZM, large systems with numerous devices can finally be obtained. Several improvements are proposed in the following works: [21] supplements some nonlinear optical effects such as two-photon absorption (TPA), free-carrier absorption (FCA), and dispersion (FCD). [15] proposes to adopt the frequency chirp-based approach instead of stepped-frequency transient analysis (SFTA) to obtain the frequency response of photonic devices more rapidly. A SPICE model based on the same modeling philosophy is also given in [25] to improve the simulation speed.

While the model has been nicely refined and developed, nonlinear interactions between different wavelengths have never been considered in the literature. The missing of these effects limits the accuracy and the scope of application of the models. To further improve the model and expand its application scenarios, some essential modifications to the previous framework are

made, and two critical nonlinear interactions, namely FWM and cross-phase modulation (XPM), are considered in this work. Other interaction effects are expected to be further included based on the proposed framework.

3. Compact Photonic Model for FWM

3.1. Baseband equivalence

To better illustrate the limitations of the previous framework, we first briefly introduce the basic idea of electronic-photonic co-simulation modeling. For EPHICs, there is a significant frequency mismatch between the optical and electrical signals: the carrier frequency of light is typically very high (~ 200 THz), while the frequencies of the modulated signals loaded on the light are about 100 GHz, and that of the thermal fluctuation is around 100 kHz. The frequency mismatch forces the simulator to use the most minor time step, resulting in an extremely long simulation time. To solve this problem, baseband equivalence is proposed in [7, 14]. By shifting the optical carrier frequency to the baseband frequency, the optical signal can be expressed as

$$\begin{aligned}\tilde{E}(z, t) &= e^{j\omega_R t} \int_{-\infty}^{\infty} \tilde{E}_{\text{bb}}(z, \Delta\omega) e^{j\Delta\omega t} d\Delta\omega = e^{j\omega_R t} \tilde{E}_{\text{bb}}(z, t) \\ &= e^{\alpha z + j(\beta(\omega_R)z + \omega_R t)} \tilde{E}_{\text{bb}}\left(0, t - \frac{zn_g}{c}\right)\end{aligned}\quad (1)$$

where the $\tilde{E}(z, t)$ is the analytic electric field, $\tilde{E}_{\text{bb}}(z, t)$ is its baseband equivalent field, they are both presented in complex numbers. Meanwhile, c is light propagation speed, ω_R is the reference frequency, and the actual frequency ω then can be written as $\omega_R + \Delta\omega$. If only considering the first-order dispersion, the wave vector β at frequency ω is given as

$$\beta(\omega) = \frac{\omega \cdot n_{\text{eff}}(\omega)}{c} = \beta(\omega_R) + \Delta\omega \left. \frac{\partial \beta}{\partial \omega} \right|_{\omega_R}, \quad (2)$$

where $n_{\text{eff}}(\omega)$ is the effective refractive index at frequency ω , and $\left. \frac{\partial \beta}{\partial \omega} \right|_{\omega_R} = \frac{n_g}{c}$, where n_g is the group index. For a waveguide of length L , the output field $\tilde{E}_{\text{bb}}(L, z)$ as a function of the input field $\tilde{E}_{\text{bb}}(0, z)$ is given by

$$\tilde{E}_{\text{bb}}(L, t) = e^{\alpha L + j\beta(\omega_R)L} \tilde{E}_{\text{bb}}\left(0, t - \frac{n_g L}{c}\right), \quad (3)$$

where $e^{\alpha L}$ is the propagation loss, $e^{j\beta(\omega_R)L}$ is phase shift with respect to ω_R , and the time delay $\frac{n_g L}{c}$ is used to present the phase shift with respect to $\Delta\omega$.

3.2. Limitations of previous framework

Fig. 1 shows the previous framework of the waveguide model. While this framework is appropriate in many applications, it cannot adequately meet the requirements of applications involving nonlinear optical interactions.

The first limitation of the previous framework is that the processing for input lights at different frequencies is identical and uncorrelated. As shown in Fig. 1, the previous framework provides only a single channel for lights transmitting in the same direction, including dual buses representing real and imaginary parts. Meanwhile, the signals at different relative frequencies are superimposed linearly in numerical form. Therefore, all processing (i.e., attenuation, phase shift, and time delay) is the same for different optical signals. Users cannot treat signals with different relative frequencies separately, and the nonlinear interactions between different wavelengths also fail to be presented.

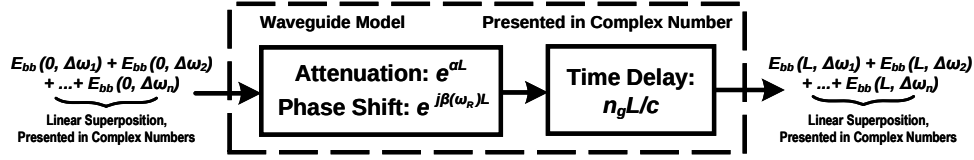


Fig. 1. The previous framework for waveguide model.

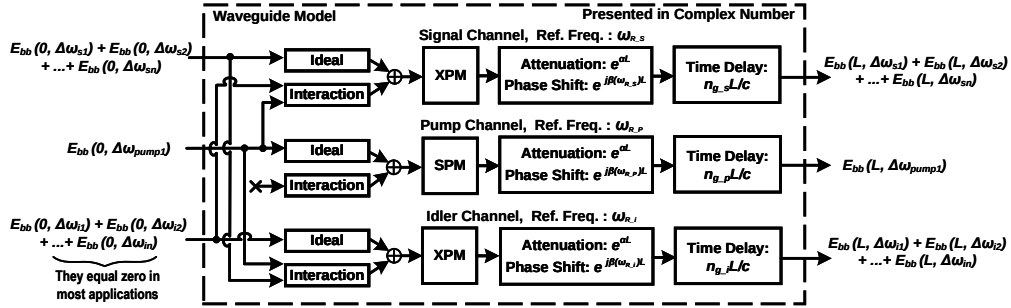


Fig. 2. The proposed framework for waveguide model.

The simulation speed is the second limitation of supporting systems that incorporate the FWM effect. For applications like wavelength conversion, the maximum wavelength difference between the signal, pump, and idler waves is usually more than a few nm (several hundred GHz or even several THz in terms of frequency). However, to ensure simulation speed, the model requires all signal frequencies to lie near the reference frequency, typically within 100 GHz. For this reason, while the previous framework can, in principle, support the simulation of systems with large frequency differences, the simulation speed will be significantly reduced, and it cannot fully satisfy our demand for rapid simulation.

3.3. Proposed framework

Fig. 2 shows the proposed framework of the waveguide model for FWM. Three different channels represent signal, pump, and idler waves, respectively. To incorporate the nonlinear optical interactions, each input of channels contains two parts, i.e., the ideal part and the nonlinear interaction part. The ideal part is the light normally incident by the laser without considering any non-ideal effects, and the output of the nonlinear interaction part is the light generated by nonlinear optical interactions like FWM. The values of the two parts will be processed separately according to the theory and then superimposed together. Meanwhile, we also use tailored rather than uniform processing for different channels. For instance, self-phase modulation (SPM) is characterized in the pump light channel, while XPM is featured in the signal and idler channels. Moreover, we use different effective and group refractive index for different waves. The tailored processing can help improve the accuracy of simulation. We also adopt three different reference wavelengths for the signal, pump, and idler waves to avoid long simulation time due to large wavelength differences. The actual wavelengths of three kinds of light can be chosen randomly around their corresponding reference wavelength. They can also freely switch to different reference wavelengths if necessary for other applications.

Note that the nonlinear interactions depend mainly on the high-order nonlinear coefficient of the medium, as well as the optical power of the laser. Since the power of the signals in FWM

is usually much weaker than that of the pump light, the influence of the interactions between different signals is commonly negligible and can be ignored under most conditions. For this reason, we mainly consider the interactions between pump and signal waves in this program. However, the proposed framework is scalable. A finer analysis can be expected by separating different frequency signals with FFT, adding interaction parts to them, and finally completing the distinctive parts of all processing.

3.4. Model description

In this part, we illustrate the principle and description of the proposed model based on degenerate four-wave mixing (DFWM). The version for non-degenerate four-wave mixing (NDFWM) can be easily obtained in the same way. For DFWM, the frequency relationship between the pump, signal, and idler waves is given by

$$\omega_s + \omega_i = 2\omega_p, \quad (4)$$

where $\omega_{p,s,i}$ are the actual frequencies of the pump, signal, and idler waves, respectively. Since the relationship between actual frequency ω , relative frequency $\Delta\omega$ and reference frequency ω_R is $\omega = \omega_R + \Delta\omega$, the relative frequencies of the pump, signal and idler waves could still satisfy $\Delta\omega_i = 2\Delta\omega_p - \Delta\omega_s$ after being shifted to the baseband if the relationship between reference frequencies is defined as $\omega_{R_i} = 2\omega_{R_p} - \omega_{R_s}$.

Considering the linear transmission loss, two-photon absorption (TPA), free-carrier absorption (FCA), and dispersion (FCD), XPM and SPM in waveguide, the coupled equations of DFWM for small-signal analysis, i.e., $|E_p| \gg |E_s|, |E_i|$, can be written as [34]

$$\frac{\partial \tilde{E}_p}{\partial z} = -\frac{1}{2}\alpha' \tilde{E}_p + i(\beta(\omega_p) + \gamma P_p) \tilde{E}_p, \quad (5)$$

$$\frac{\partial \tilde{E}_s}{\partial z} = -\frac{1}{2}\alpha' \tilde{E}_s + i(\beta(\omega_s) + 2\gamma P_p) \tilde{E}_s + i\gamma \tilde{E}_p \tilde{E}_p \tilde{E}_s^*, \quad (6)$$

$$\frac{\partial \tilde{E}_i}{\partial z} = -\frac{1}{2}\alpha' \tilde{E}_i + i(\beta(\omega_i) + 2\gamma P_p) \tilde{E}_i + i\gamma \tilde{E}_p \tilde{E}_p \tilde{E}_s^*, \quad (7)$$

where $\tilde{E}_{p,s,i}$ are the electrical fields of the pump, signal, and idler waves, respectively. $\alpha' = \alpha + \alpha_{\text{TPA}} + \alpha_{\text{FCA}}$, where α , α_{TPA} , and α_{FCA} are the linear transmission loss coefficient, as well as the nonlinear transmission loss coefficients caused by TPA and FCA, respectively. Moreover, $\beta(\omega_{p,s,i})$ are the wave vectors, γ is the nonlinear parameter, and P_p is the input pump light power. Notice that the equations (5)-(7) still holds under the baseband equivalence if we define $\omega_{R_s} + \omega_{R_i} = 2\omega_{R_p}$.

For low power excitation, parametric gain is negligible, and the interaction part of the signal in our framework equals zero. The coupled equations (5)-(7) after baseband equivalence can be further simplified and their analytical solutions are given by [35]

$$\tilde{E}_{\text{bb}_p}(L, t) = \tilde{E}_{\text{bb}_p}(0, t) \exp \left[-\alpha' L/2 + i(\beta(\omega_p) + \gamma P_p) L \right], \quad (8)$$

$$\tilde{E}_{\text{bb}_s}(L, t) = \tilde{E}_{\text{bb}_s}(0, t) \exp \left[-\alpha' L/2 + i(\beta(\omega_s) + 2\gamma P_p) L \right], \quad (9)$$

$$\tilde{E}_{\text{bb}_i}(L, t) = \left(\tilde{E}_{\text{bb}_i}(0, t) + i\eta\gamma \tilde{E}_{\text{bb}_p}^2(0, t) \tilde{E}_{\text{bb}_s}^*(0, t) \right) \exp \left[-\alpha' L/2 + i(\beta(\omega_i) + 2\gamma P_p) L \right], \quad (10)$$

where η is the efficiency factor affected by phase mismatch, given by $\eta = \frac{1 - \exp(-\alpha' L + i\Delta k L)}{\alpha' - i\Delta k}$. Δk is the total phase mismatch, which equals $\Delta\beta + 2\gamma P_p$. $\Delta\beta = \beta_s + \beta_i - 2\beta_p$ accounts for the linear phase mismatch of three involved waves and is related to the group velocity dispersion β_2 as $\Delta\beta = \beta_2(\omega_s - \omega_p)^2$ if higher order dispersions are neglected.

218 For high power excitation, parametric gain can not be neglected, and the interaction part of
 219 signal must be taken into account. The analytic solutions of the coupled equations (5)-(7) are
 220 then given by [36]

$$\tilde{E}_{bb_p}(L, t) = \tilde{E}_{bb_p}(0, t) \exp \left[-\alpha' L/2 + i (\beta (\omega_p) + \gamma P_p) L \right], \quad (11)$$

$$\tilde{E}_{bb_s}(L, t) = \left(u(L) \tilde{E}_{bb_s}(0, t) + v(L) \tilde{E}_{bb_i}^*(0, t) \right) \exp \left[-\alpha' L/2 + i (\beta (\omega_s) L + \varphi_{NL}) \right], \quad (12)$$

$$\tilde{E}_{bb_i}(L, t) = \left(u(L) \tilde{E}_{bb_i}(0, t) + v(L) \tilde{E}_{bb_s}^*(0, t) \right) \exp \left[-\alpha' L/2 + i (\beta (\omega_i) L + \varphi_{NL}) \right], \quad (13)$$

223 where $u(L)$ and $v(L)$ are found as $u(L) = [\cosh(g) + iq \sinh(g)/g] \exp(-\alpha' L/2)$, and $v(L) =$
 224 $[i\gamma \tilde{E}_{bb_p}^2(0, t) L_{\text{eff}} \sinh(g)/g] \exp(-\alpha' L/2)$. Meanwhile, the gain coefficient g can be expressed
 225 as $\sqrt{(\gamma P_p L_{\text{eff}})^2 - q^2}$, and $q = \Delta\beta L/2 + \gamma P L_{\text{eff}}$. $L_{\text{eff}} = (1 - e^{-\alpha' L})/\alpha'$ represents the effective
 226 length of waveguide considering the loss, φ_{NL} is the nonlinear phase and equals $\gamma P_p L_{\text{eff}} - \Delta\beta L/2$.

227 The different terms in the equations (8)-(10) and (11)-(13) can each correspond to different
 228 parts of the proposed framework. We can further determine the expression of the ideal part
 229 and the nonlinear interaction part, as well as the description of the time delay, attenuation, and
 230 phase shift from the equations. Ultimately, we implement the Verilog-A code under the proposed
 231 framework and based on the theoretical derivation results. Notice that we can obtain two models
 232 based on equations (8)-(10) and equations (8)-(10), respectively. The model based on equations
 233 (8)-(10) is only for conditions with relatively low input pump power. The model based on
 234 equations (11)-(13) has broader scenarios. However, it is more complicated and needs relatively
 235 more time in simulation.

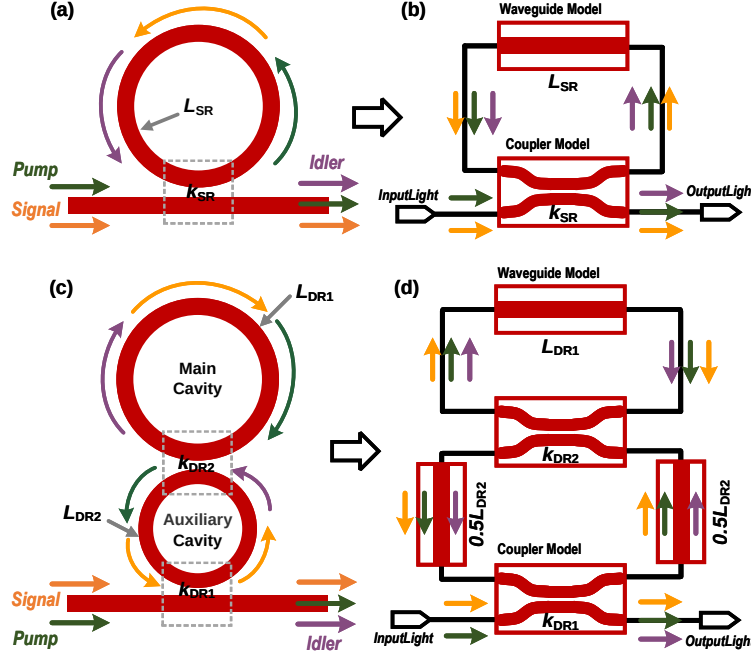


Fig. 3. The structure (a) and the schematic (b) of the single-ring model; the structure (c) and the schematic (d) of the dual-ring model

Table 2. Key Simulation Parameters for the Single and Double Micro-rings Models

Parameter	Value	Parameter	Value	Parameter	Value
Material	Si ₃ N ₄	Effective Index n_{eff}	1.808	Group Index n_g	2.17
Loss α (dB/cm)	0.2	β_2 (ps ² /km)	-5000	γ (W ⁻¹ /m ⁻¹)	1
L_{SR} (μm) (Circumference)	1200.22	L_{DR1} (μm) (Circumference)	1204.51	L_{DR2} (μm) (Circumference)	602.25
κ_{SR} (Coupling coefficient)	0.3761	κ_{DR1} (Coupling coefficient)	0.7403	κ_{DR2} (Coupling coefficient)	0.4076
P_{pump} (dBm) (For Single Ring)	20 & 30	P_{pump} (dBm) (For Double Ring)	22	P_{signal} (dBm)	0

4. Model Validation

4.1. Simulation results

Optical parametric amplifiers (OPAs) are optical amplifiers powered by FWM and have been recognized as promising candidates for high-speed optical communication systems. High-quality(Q) micro-ring resonators (MRRs) show great potential as on-chip OPAs due to the enhanced light-matter interaction. For this reason, we present two examples based on micro-ring, including a single-ring and a dual-ring structure, to demonstrate the utility and accuracy of our proposed model.

Fig. 3 (a)(b) shows the structure and the schematic of the single-ring model. As shown in Fig. 3(b), the single ring consists of a straight waveguide and a ring waveguide. We implement its Verilog-A compact model by connecting a directional coupler model and a straight waveguide model. Table. 2 presents the critical simulation parameters of the single-ring model.

Fig. 4 shows the simulated conversion efficiency as a function of frequency detuning of signal when input pump light power is 20 dBm. Meanwhile, Fig. 5 (a)(b) and Fig. 5 (c)(d) show the simulated conversion efficiency and parametric gain as a function of frequency detuning of signal when input pump light power is 33 dBm, respectively. In the case of high pump excitation, the parametric gain compensates for the transmission loss, thus enhancing the effective Q factor of the single ring [37]. The dark red dots in Fig. 4 and Fig. 5 are obtained from a numerically solved full-map coupled nonlinear Schrödinger equation [38], which tracks the nonlinear process of light in every round trip:

$$\frac{\partial A}{\partial z} = \left[-\alpha - i\delta - i\frac{\beta_2}{2} \frac{\partial^2}{\partial \tau^2} + i\gamma|A|^2 \right] A + \sum_{n=-\infty}^{+\infty} \delta(z - nL) [i\kappa A_{\text{in}} - (1 - r)A_{\text{in}}], \quad (14)$$

where A is the complex field amplitude in the ring, including pump, signal, and idler waves. δ is the detuning between the pump light and the nearest resonance peak. In the coupling region, the input-output relation follows $A_{\text{out}} = i\kappa A|_{z=nL} + rA$, and n represents the n th round-trip. The standard split-step Fourier method is numerically applied to solve (14). The numerical results are in good agreement with the results of the proposed compact model, revealing the validity of the model.

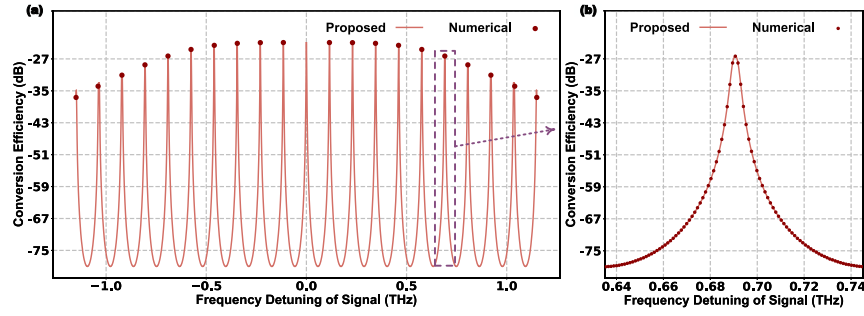


Fig. 4. Simulated conversion efficiency with proposed single micro-ring model and numerical full-map solution when input pump light power is 20 dBm.

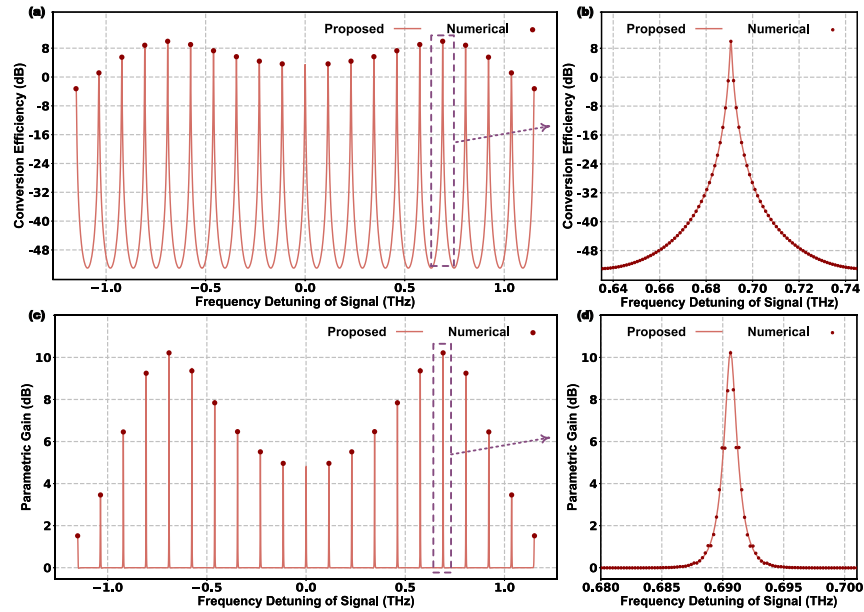


Fig. 5. Simulated conversion efficiency (a)(b) and parametric gain (c)(d) with proposed single-ring model and numerical full-map solution when input pump light power is 33 dBm.

Fig. 3 (c)(d) shows the structure and the schematic of the dual-ring model. The dual-ring structure comprises a straight waveguide and two ring waveguides, where the length of L_{DR1} is twice that of L_{DR2} . As shown in Fig. 3(d), the dual-ring model can be implemented using two coupler models and three straight waveguide models. The critical simulation parameters of the dual-ring model are also given in Table. 2.

Fig. 6 shows the simulated conversion efficiency as a function of frequency detuning of signal when input pump light power is 22 dBm. The dark red dots in Fig. 6 are also derived from the full-map coupled nonlinear Schrödinger equation, and its expression is analogous to that of the single-ring structure. The main and auxiliary cavities are described by separate coupling equations, each incorporating additional boundary coupling conditions caused by introducing new coupling nodes. The condition can be expressed as $\delta(z - nL - L/2)f(A_{\text{main}}, A_{\text{aux}})$, where $f(A_{\text{main}}, A_{\text{aux}})$ represents the electric field function in the main and auxiliary cavities. Notice that the difference between the numerical results and our simulation results at the resonance peak

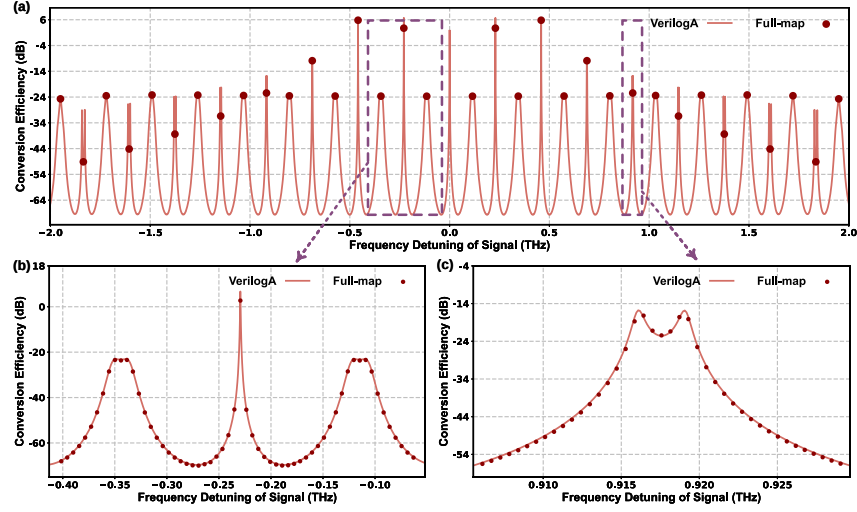


Fig. 6. Simulated conversion efficiency with proposed dual-ring model and numerical full-map solution when input pump light power is 22 dBm.

in Fig. 6(a) is not the error but the illusion due to its broad simulated transmission spectrum. As shown in localized Fig. 6(c), there is a dip at the center of the resonance peak, which is attributed to the coupling rate between the two rings significantly exceeding the intrinsic losses of the rings. The simulation results of the dual-ring model also match the numerically solved results.

Note that the proposed model is not limited to describing FWM phenomenon in the above two devices; it is a general model and can also be applied to modeling FWM phenomenon in other more complex photonic devices or even large systems.

4.2. Experiment results

The simulation results of the proposed model are also in good agreement with the experimental results reported in our previous work [37, 39]. The photonic devices used for the experiments, including a single-ring and a dual-ring structure, are also fabricated on the Si_3N_4 platform. Meanwhile, all the device parameters are consistent with the parameters in Table. 2, except for k_{SR} , which equals 0.1243.

In the single-ring experiment shown in [37], a CW pump of 21.5 dBm power and a signal of 0.5 dBm power are injected into the bus waveguide. In the on-resonance situation, the pump wave is gradually tuned from the red-detuned frequency to approach the resonant frequency. The pump wavelength is 1550.17 nm, and the signal wavelength is 1549.24 nm due to the thermal drift effect of the micro-ring. In the off-resonance situation, the pump and signal wavelengths are set directly at the same wavelength as in the on-resonance situation. The idler wave is generated at 1551.10 nm when the pump wave is on resonance, with a CE of 3.36 dB, while the signal is amplified with a gain of 4.72 dB compared with the situation of off-resonance. The simulated gain and CE are 5.36dB and 5.24 dB, slightly higher than the experiment results. The errors are likely due to process deviations and the cascaded FWM effect. They are negligible and will not affect the reliability of the electronic-photon co-simulation.

In the dual-ring experiment shown in [39], the spectrum of the dual-ring structure is manipulated by PT symmetry, manifested as an alternating bandwidth of narrow and wide resonance peaks. By injecting the pump and signal lights into the narrow and wide peaks, we can simultaneously maximize the pump efficiency and the bandwidth of wavelength conversion. Additionally, due to the intense cascaded FWM effect in the main cavity of the dual-ring structure, the pump

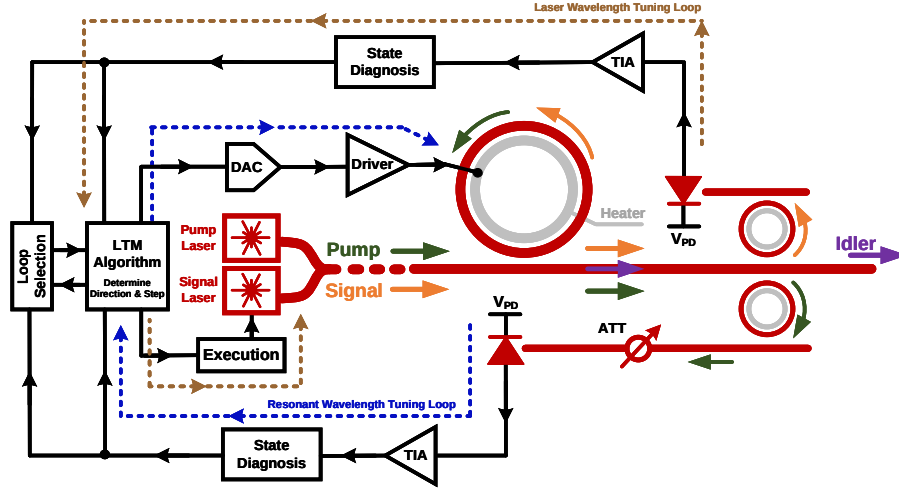


Fig. 7. The architecture of the wavelength locking circuits for wavelength conversion.

power injected into the straight waveguide needs to be maintained at a lower level. With a pump power of 4 dBm and a signal power of 5.6 dBm, we measured a conversion efficiency of -57.60 dB, slightly lower than the results of -59.09 dB obtained using the proposed model. This error also falls within an acceptable error range and has a negligible impact on the reliability of the co-simulation.

Due to the alignment challenge induced by intricate thermal effects and imperfection of the nanofabrication processes, the experiment results are only obtained at a given frequency under relatively low-power pump conditions. More carefully designed devices and high-precision closed-loop wavelength-locking circuits [40, 41] are expected to solve this problem and offer more thorough and accurate experimental results. The proposed model can help users design wavelength-locking or other attached circuits for photonic devices more efficiently.

4.3. Closed-loop co-simulation example

To further verify the effectiveness of the proposed model, an example of closed-loop wavelength locking with high-Q MRR is presented in this part. For a consistently high conversion efficiency, both the pump and signal wavelengths need to be aligned with the resonant wavelength of the micro-ring, regardless of ambient temperature variation, process deviations, and input laser fluctuations. Therefore, the closed-loop wavelength locking circuits are required to initialize and compensate for the real-time dynamic deviations. Fig. 7 shows the architecture of the wavelength-locking circuits for wavelength conversion. The architecture comprises two feedback loops, i.e., the resonant wavelength tuning loop and the laser wavelength tuning loop. The two loops are not turned on simultaneously, and the loop selection unit realizes the switch between the two loops according to the sampling information. During the initialization process, the resonant wavelength tuning loop is first turned on, the state diagnosis unit detects the operating state of the optical device based on the sampling results from TIA, and the control algorithm unit determines the tuning direction and the step according to lock-to-minimum (LTM) logic. The driver alters the ambient temperature around the MRR by adjusting the amount of power applied to the integrated heater, which, in turn, regulates the device's operating state. When the pump light wavelength is aligned with the resonant wavelength, the driver maintains a constant output power, and the loop selection unit turns on the laser wavelength tuning loop while switching off the resonant wavelength tuning loop. The alignment between the signal light wavelength and

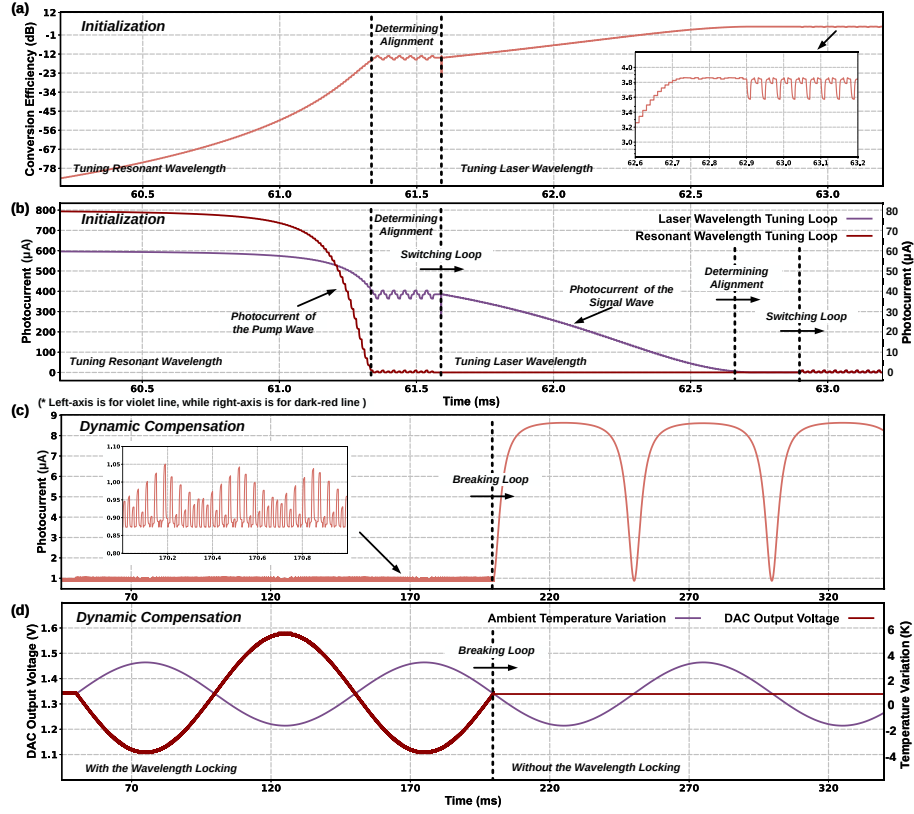


Fig. 8. The transient simulation waveforms of wavelength locking circuits in (a)(b) initialization state and in (c)(d) dynamic compensation state.

the resonant wavelength is achieved by adjusting the input laser wavelength of the signal. After completion of the initialization, the system switches back to the resonant wavelength tuning loop for real-time compensation of dynamic deviations.

Fig. 8 shows the transient simulation waveforms of wavelength-locking circuits in the initialization and dynamic compensation states, respectively. The closed-loop simulation utilizes Cadence Virtuoso (IC618) as the EDA simulation platform with Intel Xeon(R) CPU E5-2687W and 128 GB memory. In the initialization state, as shown in Fig. 8 (a)(b), the photocurrents of both loops are locked at their respective minimum values. Therefore, the conversion efficiency shifts from about -78 dB to about 3.7 dB, remaining at its maximum value. In the dynamic compensation state, as shown in Fig. 8 (c)(d), when a 10 Hz ambient temperature disturbance is applied to the MRR, the wavelength-locking circuits dynamically compensate for the ambient temperature variation and keep the photocurrent to its minimum. After breaking the loop, the photocurrent starts to vary over a relatively large range.

The idler light generated in our system can be further processed in subsequent systems, thus forming a more complete and complex NOSP system, which is impossible with previous models. Meanwhile, since the proposed model could provide adequate and accurate simulation results, it can help users to efficiently design and validate better architectures, algorithms, and circuit structures for different EPHIC systems.

5. Conclusion

In this paper, we demonstrate a Verilog-A compact photonic model based on coupled-mode theory for four-wave mixing (FWM) to present a general framework and methodology for modeling nonlinear interactions in electronic-photonic co-simulation. The model provides accurate and efficient simulation results for the design of the EPHIC systems incorporating FWM. As a general model, it can assist users in designing attached circuits for various photonic devices, addressing the challenge of traditional co-simulation models failing in the presence of nonlinear interactions. Simulation results derived from our models agree with the results of the numerically solved full-map Schrödinger equation and experimental results. A closed-loop electronic-photonic co-simulation example is also presented. The models for other optical nonlinear interactions can also be expected based on the proposed framework. Moreover, the model based on our framework can also be implemented in SPICE to further improve the simulation speed.

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